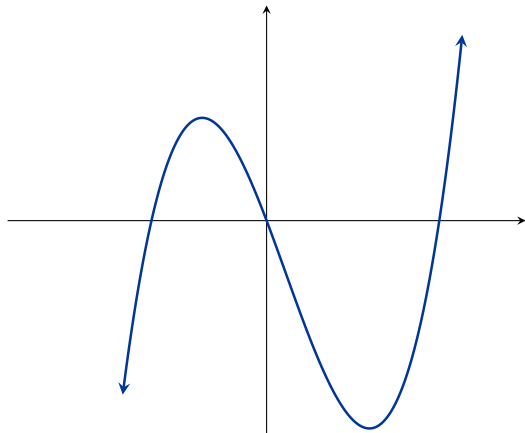


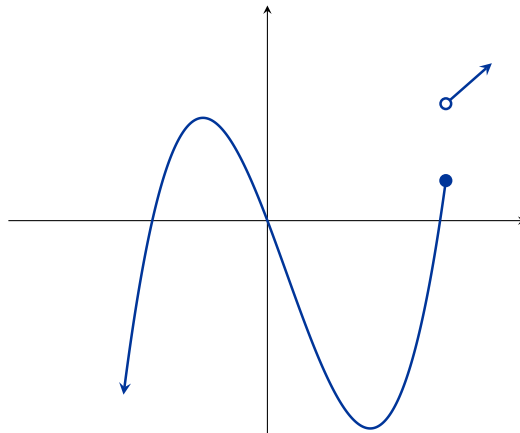
3.4: Graphs of Polynomial Functions

Graphs of polynomials are smooth and continuous with infinite end behavior.

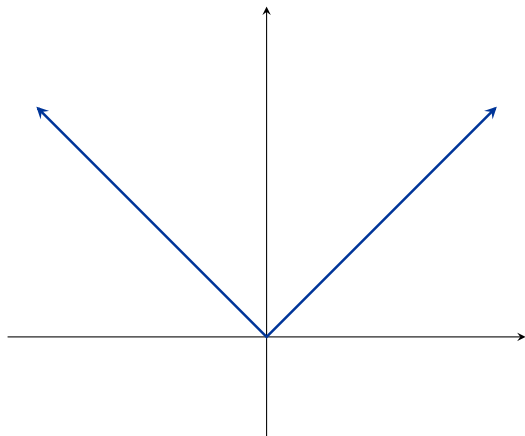
Smooth and continuous



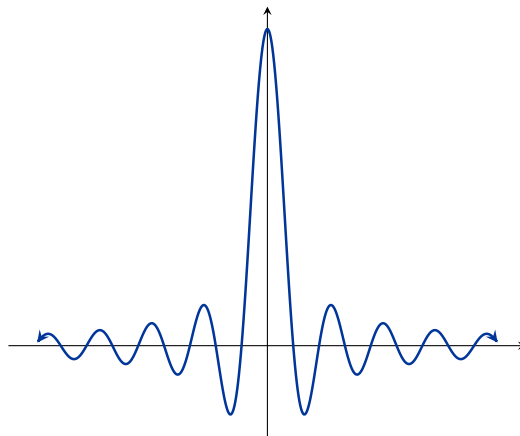
Not continuous



Not smooth



Finite end behavior



Example. Find the x - and y -intercepts of the following functions:

$$f(x) = 2(x - 4)(x + 1)(x - 6)$$

$$g(x) = 4x(x - 2)^2(x + 1)$$

$$h(x) = x^4 - x^2$$

$$j(x) = x^3 + 2x^2 - 9x - 18$$

Definition. (Multiplicity of a Root)

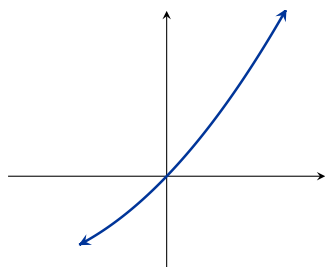
If a polynomial contains a factor of the form $(x - h)^p$, then the behavior near the x -intercept h is determined by the power p . We say that $x = h$ is a zero of **multiplicity** p .

When p is,

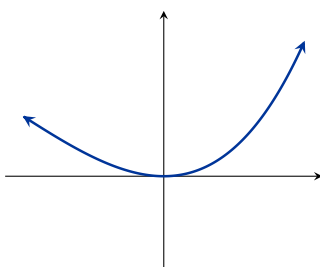
- even, the graph will touch the x -axis
- odd, the graph will cross the x -axis

The degree of the polynomial is the sum of the multiplicities.

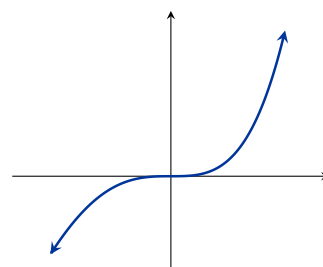
Single zero



Zero with multiplicity 2



Zero with multiplicity 3



Example. For the following, find the zeros, and the multiplicities of each:

$$f(x) = (x + 2)^3(x - 3)^2(x^2 - 1)$$

$$g(x) = x^5(x - 1)^3(x + 2)$$

Example. Find a formula for the polynomial of minimum degree represented in the graph below:

